

**A REPORT
ON
AI GPU SUMMER INTERNSHIP**

Accelerated by

NVIDIA

Submitted by

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Under the guidance of,

Mr.Senthil Kumar H

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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

**COMPUTER SCIENCE AND ENGINEERING
(COMPUTER ENGINEERING)**

AT



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PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the report of **CSE7000 – Internship** on “**AI GPU SUMMER INTERNSHIP**” being submitted by “**RISHIKA BS**” bearing roll number “**20241CSE0764**” in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Engineering (Computer Science and Engineering)** is a bonafide work carried out under our supervision.

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DECLARATION

I hereby declare that the work, which is being presented in the report entitled “**AI GPU SUMMER INTERNSHIP**” in partial fulfillment for the award of Degree of **Bachelor of Technology in Computer Science and Engineering (Computer Science and Engineering)**, is a record of my own investigations carried under the guidance of **Mr. Senthil Kumar H**, Assistant Professor and **Ms. Vineetha B**, Assistant Professor, Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

I have not submitted the matter presented in this report anywhere for the award of any other Degree.

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INTERNSHIP COMPLETION CERTIFICATE

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ABSTRACT

Diabetes is one of the most prevalent chronic diseases worldwide and has become a major public health concern due to its increasing incidence and long-term health complications. The disease occurs when the body is unable to effectively regulate blood glucose levels, leading to serious medical conditions such as heart disease, kidney failure, nerve damage, and vision impairment. Early diagnosis and timely treatment are essential to reduce the risk of these complications and improve the quality of life of patients. However, traditional diagnostic procedures often require laboratory testing and medical consultation, which may not always be readily available.

The Diabetes Prediction System is developed to assist in the early detection of diabetes using Machine Learning techniques. The project aims to predict whether a person is likely to have diabetes based on several medical parameters, including glucose level, blood pressure, insulin level, skin thickness, body mass index (BMI), age, number of pregnancies, and diabetes pedigree function. These parameters are analyzed using a trained machine learning model to provide accurate and reliable predictions.

The system utilizes the PIMA Indians Diabetes Dataset, a widely recognized dataset used for diabetes prediction research. Before training the model, the dataset undergoes preprocessing steps such as data cleaning, validation, feature selection, and data splitting. These processes ensure that the data is suitable for machine learning analysis and improves the overall performance of the predictive model.

A Logistic Regression algorithm is selected for this project because it is efficient, easy to interpret, and highly suitable for binary classification problems. The model is trained using 80% of the dataset and tested on the remaining 20% to evaluate its performance. Various evaluation metrics such as Accuracy, Precision, Recall, and F1 Score are used to measure the effectiveness of the model.

To make the system accessible and user-friendly, a web application is developed using Streamlit. The application allows users to enter their medical information through an interactive interface and instantly receive a prediction indicating whether they are likely to be diabetic or non-diabetic. This approach demonstrates how Artificial Intelligence and Machine Learning can be applied in healthcare to support decision-making and promote preventive healthcare practices.

1.INTRODUCTION

Diabetes mellitus is a chronic metabolic disorder that affects millions of people around the world. It occurs when the body either does not produce enough insulin or cannot effectively use the insulin it produces. As a result, blood glucose levels rise beyond normal limits, leading to various health complications if left untreated. According to healthcare organizations worldwide, the number of diabetes cases has increased significantly over the past few decades due to lifestyle changes, poor dietary habits, obesity, lack of physical activity, and genetic factors.

Early detection of diabetes plays a crucial role in preventing severe complications such as cardiovascular diseases, kidney failure, nerve damage, blindness, and other health issues. Conventional methods of diabetes diagnosis involve laboratory tests and medical examinations, which may be time-consuming and costly. With advancements in Artificial Intelligence and Machine Learning, predictive healthcare systems can now assist medical professionals in identifying patients who may be at risk of developing diabetes.

Machine Learning enables computers to learn patterns from historical data and make predictions on new data. In healthcare, machine learning techniques are increasingly being used to support disease diagnosis, treatment planning, and patient monitoring. By analyzing patient health records and identifying hidden relationships among medical parameters, machine learning models can provide valuable insights for early disease detection.

The Diabetes Prediction System developed in this project uses the PIMA Indians Diabetes Dataset to train a Logistic Regression model. The dataset contains various medical attributes such as glucose concentration, blood pressure, skin thickness, insulin level, body mass index (BMI), diabetes pedigree function, age, and number of pregnancies. These features are used to determine the likelihood of diabetes in a patient.

The project also includes a Streamlit-based web application that provides a simple and interactive user interface. Users can enter their health-related information, and the trained machine learning model generates an instant prediction regarding their diabetes status. The application is designed to be easy to use, making healthcare prediction more accessible to users.

The primary goal of this project is to demonstrate the practical application of machine learning in healthcare and to develop a reliable system that can assist in the early identification of diabetes risk. The system not only improves awareness about diabetes but also highlights the potential of artificial intelligence in modern healthcare solutions.

2.OBJECTIVE

The primary objective of the Diabetes Prediction System is to develop an intelligent healthcare application capable of predicting the likelihood of diabetes using Machine Learning techniques. Diabetes is a rapidly growing health concern worldwide, and early detection can significantly reduce the risk of severe complications. Therefore, the project aims to provide a simple, efficient, and reliable prediction system that can assist users in understanding their diabetes risk at an early stage.

The project focuses on utilizing the PIMA Indians Diabetes Dataset, which contains several medical parameters related to diabetes diagnosis. By analyzing features such as glucose level, blood pressure, insulin level, skin thickness, body mass index (BMI), diabetes pedigree function, age, and number of pregnancies, the system predicts whether a patient is diabetic or non-diabetic.

Another objective of the project is to apply Machine Learning concepts in a real-world healthcare application. Logistic Regression is selected as the prediction algorithm because of its effectiveness in binary classification problems and its ability to produce interpretable results. The model is trained using historical medical data and tested to ensure accurate prediction performance.

The project also aims to develop a user-friendly web application using Streamlit. Through this interface, users can easily enter their health information and receive instant prediction results. The system eliminates the need for complex technical knowledge and makes healthcare prediction accessible to a broader audience.

In addition, the project seeks to evaluate the model using standard performance metrics such as Accuracy, Precision, Recall, and F1 Score. These metrics help determine the reliability and effectiveness of the prediction system. The overall objective is to demonstrate how Artificial Intelligence and Machine Learning can contribute to healthcare by supporting early diagnosis, improving awareness, and promoting preventive medical care.

3.WORK PROGRESS

The development of the Diabetes Prediction System was carried out through a series of systematic phases. The first phase involved understanding the problem statement and studying diabetes-related medical parameters that influence disease prediction. Research was conducted on various machine learning techniques and healthcare prediction systems to determine the most suitable approach for the project.

The next phase focused on dataset collection and analysis. The PIMA Indians Diabetes Dataset was selected because it is widely used in diabetes prediction research and contains relevant medical attributes. The dataset consists of 768 patient records and includes features such as glucose level, blood pressure, insulin, BMI, age, pregnancies, skin thickness, and diabetes pedigree function.

After collecting the dataset, preprocessing activities were performed. The data was loaded using the Pandas library and carefully examined for inconsistencies. Feature selection was carried out to identify the most relevant attributes for prediction. The dataset was then divided into training and testing sets, ensuring that the model could be evaluated effectively.

In the model development phase, Logistic Regression was selected and implemented using Scikit-learn. The algorithm was trained using 80% of the dataset and tested on the remaining 20%. Multiple experiments were conducted to evaluate the model's performance and ensure reliable predictions.

Following model development, a Streamlit web application was created to provide an interactive interface. Users can input their medical information through the web application, and the trained model generates instant predictions. The interface was designed to be simple, user-friendly, and accessible.

Finally, the system was tested using various input combinations to verify prediction accuracy and functionality. Performance metrics such as Accuracy, Precision, Recall, and F1 Score were calculated to assess the effectiveness of the model. The successful completion of these phases resulted in a functional diabetes prediction system capable of assisting users in assessing diabetes risk.

4.RESULTS

The Diabetes Prediction System was successfully implemented and tested using the PIMA Indians Diabetes Dataset. The Logistic Regression model demonstrated strong performance in predicting whether a patient is diabetic or non-diabetic based on various medical parameters. The experimental results indicate that machine learning techniques can effectively assist in healthcare prediction and decision-making.

To evaluate the performance of the model, standard machine learning metrics were calculated. The model achieved an accuracy of 85.06%, indicating that it correctly classified approximately 85 out of every 100 patient records. This level of accuracy demonstrates the model's ability to make reliable predictions using the available medical data.

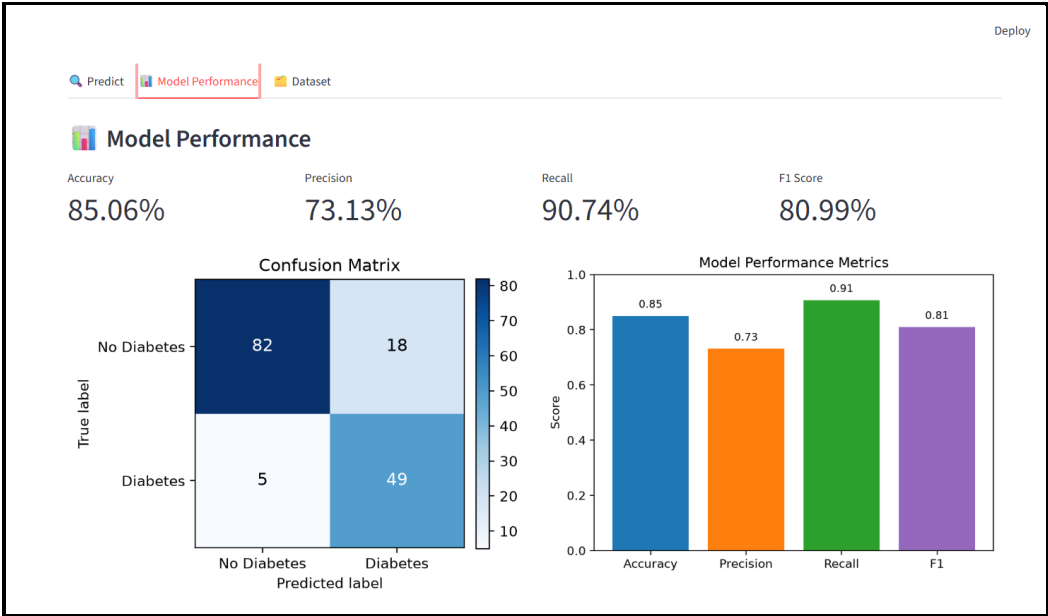
The screenshot displays the 'Diabetes Prediction System' web application. On the left, a 'Patient Data Entry' sidebar contains input fields for various medical parameters: Pregnancies (0), Glucose (mg/dL) (83), Blood Pressure (mm Hg) (90), Skin Thickness (mm) (20), Insulin (mu U/mL) (75), BMI (25.00), and Diabetes Pedigree Function (0.47). The main content area features the system's title, a subtitle, and navigation tabs for 'Predict', 'Model Performance', and 'Dataset'. The 'Predict' tab is active, showing a 'Prediction Result' section with a green checkmark and the text 'Likely Not Diabetic', along with an 'Estimated probability of diabetes: 4.3%'. To the right, a bar chart shows the 'Probability: Diabetic' at 4.3% and 'Probability: Not Diabetic' at 95.7%. A disclaimer at the bottom states: 'This tool is for educational/demo purposes only and is not a substitute for professional medical diagnosis.'

The precision value of 73.13% reflects the reliability of positive diabetes predictions. It indicates that a significant proportion of patients predicted as diabetic were correctly identified. The recall value of 90.74% shows the model's effectiveness in detecting diabetic patients. A high recall score is particularly important in healthcare applications because failing to identify a diabetic patient could lead to serious health consequences.

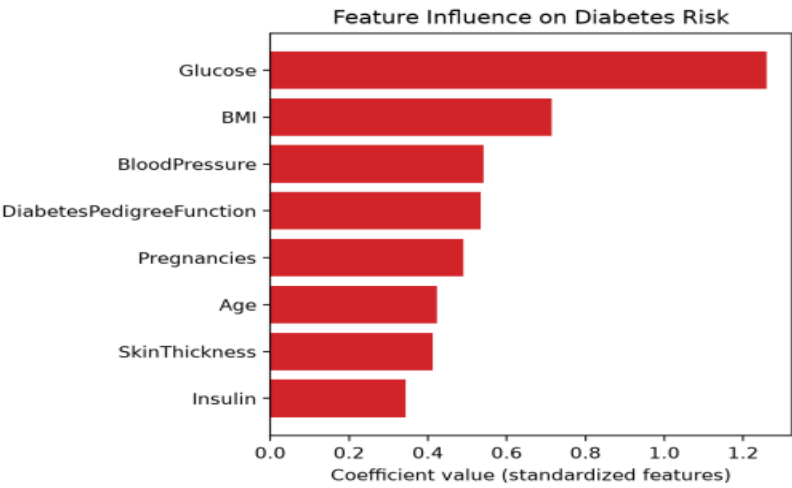
The F1 Score of 80.99% provides a balanced measure of precision and recall. The strong F1 Score confirms that the model maintains a good balance between identifying diabetic patients and minimizing incorrect predictions.

The confusion matrix further illustrates the model's performance. The model achieved 82

True Negatives, 49 True Positives, 18 False Positives, and only 5 False Negatives. These results indicate that the system successfully identifies most diabetic patients while maintaining a relatively low error rate.



Feature importance analysis revealed that Glucose Level is the most influential factor in predicting diabetes. Other significant features include BMI, Blood Pressure, Diabetes Pedigree Function, Pregnancies, Age, Skin Thickness, and Insulin Level. The findings align with medical knowledge regarding the primary risk factors associated with diabetes.



Overall, the results demonstrate that the developed system is capable of providing accurate and reliable diabetes predictions and can serve as an effective educational research-oriented health care application.

5.CONCLUSION

The Diabetes Prediction System successfully demonstrates the application of machine learning in healthcare. By utilizing the PIMA Indians Diabetes Dataset and Logistic Regression algorithm, the system achieved an accuracy of 85.06%.

The developed Streamlit application provides a simple and interactive platform for users to enter medical parameters and receive immediate diabetes risk predictions. The high recall score indicates that the model is effective in identifying diabetic patients, making it a useful educational and research-oriented healthcare application.

Future improvements may include using larger datasets, applying advanced machine learning algorithms such as Random Forest and XGBoost, and integrating real-time healthcare data to further improve prediction accuracy.

6.REFERENCES

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